

Unit 3: Overview of Business Data Journey

Assignment 6: Data Pre-Processing using Python for Business Data Analytics

Topics Covered: Introduction to Scikit-Learn; Loading Dataset; Duplication Checking, Missing Value Calculation; Data Pre-Processing - Outlier Detection, Data Cleaning; Data Transformation (Drop / Replace/ Imputation, Categorical to Numerical Conversion, Data Normalization / Standardization), Data Reduction/Feature Selection, Selection of Independent and Dependent Variable

The assignment aims to provide students a comprehensive guide on importing and pre-processing data using Python for Business Data Analytics. It covers key aspects of data pre-processing, including data cleaning, integration, transformation, and feature selection.

Scikit-Learn

Machine Learning Application: Implement a simple machine learning model (e.g., regression, classification) using Scikit-Learn on business data (e.g., customer segmentation, sales prediction)

Model Evaluation: Evaluate the model's performance using Scikit-Learn metrics (e.g., accuracy)

Data Import Using Pandas

Dataset Selection: Choose a business-related dataset (e.g., sales data, customer information, market trends, etc.) available in CSV, Excel, or any another suitable format.

Importing Data: Use the Pandas library in Python to read and import the chosen dataset into a Pandas Data Frame.

Data Cleaning

Data Inspection: Perform an initial exploration of the dataset: view first few rows, check data types, and understand column names.

Handling Missing Values: Identify missing values and implement strategies (e.g., imputation, removal) to deal with them effectively.

Handling Duplicates: Detect and remove any duplicate entries present in the dataset.

Data Consistency: Address inconsistencies in data formatting, such as standardizing date formats, handling categorical inconsistencies, etc.

Outlier Detection and Treatment: Identify outliers and apply appropriate methods (e.g., trimming, transformation) to handle them.

Data Transformation

Encoding Categorical Variables: Utilize encoding methods like one-hot encoding or label encoding to transform categorical variables into numerical format.

Normalization/Standardization: Implement normalization or scaling techniques for numerical features if required (using Scikit-learn or Pandas). Normalization is preferred when features have different ranges and the model relies on distance calculations, such as k-NN, K-means as it scales values to a fixed range (0–1). Standardization is preferred when data follows a normal distribution or when using models like Linear Regression, Logistic Regression, SVM, as it centers data around mean 0 with unit variance.

Feature Engineering: Create new features or modify existing ones to enhance the dataset's predictive power or descriptive capabilities.

Data Reduction

Dimensionality Reduction: Use scikit-learn to reduce the dataset's dimensions.

Feature Selection: Use methods like feature importance or correlation analysis to select relevant features for analysis.

Students are required to create a folder as FourDigitsRollno_Roomno_DeptCode_BDA (For Example, 0012_14_COMA_BDA) on their desktop. All the files related to the assignment should be saved within this folder. Students are required to use either Spyder or Jupyter Notebook to complete the assignment. The final working code should be saved as AssignmentNo_Rollno.py

1. A company aims to analyze customer purchase behavior for a particular product using demographic attributes such as country, age, and salary. Before applying machine learning models, the dataset

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must be cleaned, transformed, normalized, and standardized to ensure reliable and accurate analysis.

Create a simple dataset (data4. csv) according to the following structure that contains information about customers who have purchased a particular product from a company. Here, country, age, and salary are the independent variables and purchased column is the dependent variable.

Country	Age	Salary	Purchased
France	44	72000	No
Spain	27	48000	Yes
Germany	30	54000	No
Spain	38	61000	No
Germany	40		Yes
France	35	58000	Yes
Spain		52000	No
France	48	79000	Yes
Germany	50	83000	No
France	37	67000	Yes
Spain	38	61000	No

The above dataset is not in a ready-to-use state (not formatted properly, containing missing or null values). Do the following (data pre-processing) before considering the above dataset for analysis.

Importing Libraries and Dataset

- Import required libraries: NumPy, Pandas, Matplotlib.
- Load the dataset using read_csv().

Data Inspection

- Display the first 8 and last 8 rows.
- Print dataset information.
- Display the shape of the dataset.

Data Cleaning

- Identify and remove duplicate rows, if any.

Missing Value Analysis

- Display the number of missing values per column.
- Display the total number of missing values.

There are two missing values in this dataset - one in the Salary column (5th Row) and another in the Age column (7th Row)

Handling Missing Values

- Handle missing values using one of the following methods:
 - Use location-based imputation to fill missing values with a value (Your Rollno).
 - Fill missing values with mean/median.

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- Drop the rows having missing values.
- **Impute using SimpleImputer class from the scikit-learn library.**

Data Transformation

- k. Display unique values in categorical columns.
- l. Count the number of unique categories.
- m. Encode categorical variables:
 - Country: OneHotEncoder with ColumnTransformer
 - Purchased: LabelEncoder
 - Map categorical variables to numerical values using map() function.

Feature Scaling (Normalization and Standardization)

Normalization:

- n. Normalize numerical features using MinMaxScaler from scikit-learn.

Standardization:

- o. Standardize numerical features using StandardScaler from scikit-learn.

Data Reduction and Splitting

- p. Drop irrelevant columns or define independent (features) and dependent (target) variables.
- q. Split the dataset into training (80%) and testing (20%) sets using train_test_split().

9. Output Display

- r. Display the transformed dataset and train-test split outputs.
